# Major Project Synopsis

## 1. Team Details:

| Group No. | Sr. No. | Team Member | Supervisor(s) |
|---|---|---|---|
| G19 | 1. | Pranav Malhotra<br>22105036 | Dr. Jasbir Kaur |
| | 2. | Ishan Gupta<br>22105031 | |
| | 3. | Pranav Chaudhary<br>22105014 | |

## 2. Title of the project:

IoT-Enabled Edge Cardiac Monitoring System with Explainable Risk Analytics

☐ Software ☐ Hardware ☑ Hardware + Software

## 3. Objectives:

- To design and implement an ESP32-based multi-sensor cardiac monitoring system that acquires real-time physiological signals such as heart rate, $SpO_2$, and motion activity using biomedical sensors.
- To develop an edge-level signal processing mechanism that applies digital filtering techniques and extracts meaningful features (e.g., activity index) for reliable cardiac analysis.
- To build a wireless IoT communication framework that transmits processed sensor data to a server for continuous monitoring and remote accessibility.
- To create an Explainable AI-based cardiac risk prediction model that not only classifies risk levels but also provides interpretable insights into contributing health factors.
- To design a lightweight web-based dashboard and embedded alert system (OLED display and buzzer) that visualizes live vitals, risk status, and explanation graphs in a user-friendly interface.

## 4. Proposed Methodology:

- **Sensing & Data Acquisition**

  MAX30102 will capture heart rate and $SpO_2$ from the fingertip, while MPU6050 will measure motion/activity. An optional DS18B20 sensor may record skin temperature. All sensors will interface with ESP32.

- **Signal Processing & Edge Features**
  ESP32 will apply basic filtering (moving average, low-pass) to reduce noise and extract features such as heart rate, HRV, and activity level.
- **IoT Communication**
  Processed data will be sent via Wi-Fi (HTTP/MQTT) to a backend server using structured data packets for real-time monitoring.
- **Explainable AI Prediction**
  A deep learning model will classify cardiac risk levels, and Explainable AI (SHAP/LIME) will show key contributing factors.
- **Dashboard & Alerts**
  A web dashboard will display live vitals and risk level. OLED display and buzzer will provide local visual and audio alerts.
- **Testing & Validation**
  System will be tested under different activity conditions to check signal accuracy and prediction performance.
- **Manual Control**
  A push-button will allow mode switching and system reset during operation.

## 5. Tentative Hardware required:

| S.No | Component | Tentative Quantity | Tentative Cost | Justification |
|---|---|---|---|---|
| 1 | ESP32 Dev Module | 1 | Rs. 350–450 | Main controller with WiFi for data processing and communication |
| 2 | MAX30102 Heart Rate & $SpO_2$ Sensor | 1 | Rs. 250–350 | Captures PPG signals from fingertip for cardiac monitoring |
| 3 | MPU6050 Motion Sensor | 1 | Rs. 100–150 | Measures activity level and reduces motion artifacts |
| 4 | DS18B20 Temperature Sensor (Optional) | 1 | Rs. 100 | Records skin/body temperature for additional health data |
| 5 | SSD1306 OLED Display | 1 | Rs. 200–250 | Displays live vitals and risk level locally |
| 6 | Active Buzzer Module | 1 | Rs. 50–80 | Provides alert when high cardiac risk is detected |
| 7 | Push Button Switch | 1 | Rs. 20–40 | Allows manual start/reset or mode selection |
| 8 | Breadboard, Jumper Wires, Connectors | 1 set | Rs. 150–200 | Circuit connections and prototyping |
| 9 | Power Supply (5V Adapter / Battery) | 1 | Rs. 150 | Provides stable power to ESP32 and sensors |
| 10 | Enclosure / Casing | 1 | Rs. 200–300 | Protects and organizes hardware components |

## 6. Software:

- Arduino IDE will be used for ESP32 programming and sensor interfacing, while Python (TensorFlow/Scikit-learn) will be used to develop the cardiac risk prediction and Explainable AI model.
- A web-based dashboard will be developed using HTML/CSS/JavaScript or React, and a lightweight backend (FastAPI/Node.js) will handle data processing and visualization.

## 7. Block Diagram

```
                ┌──────────────────────────┐
                │        Human Body        |
                └──────────────────────────┘
                            |
                            |
        ┌───────────────────|────────────────────┐
        |                   |                    |
  MAX30102 Sensor     MPU6050 Sensor      DS18B20 (Temp)
 (Heart Rate + SpO2)    (Motion/Activity)    (Optional)
        |                   |                    |
        └──────────────── I2C / GPIO ────────────┘
                            |
                        ┌───▼───┐
                        |  ESP32  |
                        | Edge MCU |
                        └───────┘
                            |
        ┌──────────── Signal Filtering + Feature Extraction ────────────┐
                            |
                        WiFi Data
                            |
                      ┌──────▼──────┐
                      | AI Model +  |
                      | Explainable |
                      | Analytics   |
                      └──────────┘
                            |
                ┌──────────────────────────┐
                |                          |
          OLED Display              Web Dashboard
            (Local)                   (React)
                |
          Buzzer Alert
```

## 8. Timelines and Milestones:

| Phase | Duration | Description of Activities |
|---|---|---|
| **Phase 1** | Week 1 – Week 2 | Identification of problem statement, requirement analysis, literature survey on cardiac monitoring and Explainable AI, and finalization of project objectives. |
| **Phase 2** | Week 3 – Week 4 | System architecture design, preparation of block diagrams and flowcharts, selection of sensors (MAX30102, MPU6050) and software tools, and procurement of components. |
| **Phase 3** | Week 5 – Week 6 | Interfacing of MAX30102 heart rate sensor and MPU6050 motion sensor with ESP32, basic signal acquisition, and hardware setup testing. |
| **Phase 4** | Week 7 – Week 8 | Implementation of signal filtering, feature extraction (HRV, activity level), OLED display interfacing, and buzzer alert mechanism. |
| **Phase 5** | Week 9 – Week 10 | Development of AI-based cardiac risk prediction model and integration of Explainable AI techniques, along with dashboard development for real-time visualization. |
| **Phase 6** | Week 11 – Week 12 | Final system integration, testing under different activity conditions, performance validation, result analysis, report preparation, and project presentation. |

Students' Signature

Date:

Supervisor(s) Signature

Date: